

Hype Live: Biometric-based Sensory Feedback for Improving the Sense of Unity in VR Live Performance

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ABSTRACT

We propose Hype Live, a system to improve the sense of unity by sharing responses through visual, auditory, and haptic stimuli based on the biometrics of the participants in VR live performances. In this field, the sharing of reactions among the participants is one of the most important factors in improving the sense of unity. However, not many past studies have provided feedback to the participants' senses. Therefore, as a prototype of Hype Live, we used a vibration device to provide haptic feedback and recreated moshing scenes where the participants violently collide with each other like a real live performance.

Index Terms: Human-centered computing—Human computer interaction(HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

In recent years, live performances in virtual reality (VR) environments with remote participation have become free from the constraints of physical distance and time, and allow audiences to interact with each other. In real live performances, there is a sense of unity due to the unique interaction between the audiences and the performers, but in remote participatory live performances such as VR live performances, there is a problem of decreased sense of unity due to the lack of reaction sharing [3]. In order to solve this problem, there have been studies on computer-assisted liveliness promotion [4] and motion control of Non-player character (NPC) audiences [5] during live performances in the past. However, research on VR live performances to improve the sense of unity has focused on interaction through movement [1], but there is no system that aims to improve the sense of unity through reaction sharing and interaction with NPC audiences through visual, auditory, and haptic stimulation based on the biometrics of the participants. There is no such system. In this study, we propose a system called Hype Live that can improve the sense of unity by providing sensory feedback based on biometrics in a live VR environment. As a first step toward the realization of Hype Live, we implemented a moshing function and conducted preliminary experiments to determine whether haptic stimuli such as simulated moshing, which has not been verified in previous studies, would promote a sense of unity.

2 HYPE LIVE

2.1 System Overview

Figure 1 shows the real environment and the VR live environment during the use of the proposed system, Hype Live. Figure 2 shows

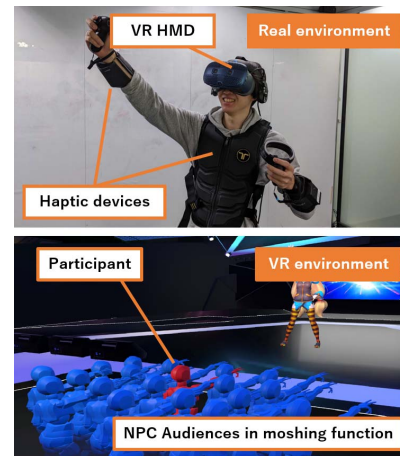


Figure 1: Participant in a real environment and NPC audiences moshing in a VR environment.

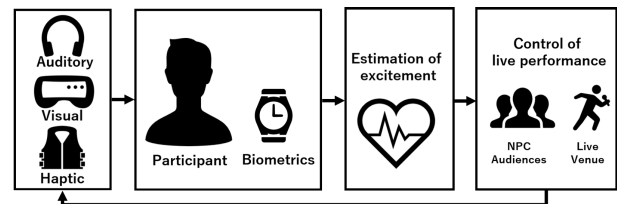


Figure 2: Hype Live system configuration

the system configuration of Hype Live. There are two main functions to support the improvement of the sense of unity: one is the biometric-based estimation of excitement, and another is the interaction function with the NPC audiences. Participants wear a wristwatch-type wearable device that can measure their pulse and obtain biometrics to estimate the timing and magnitude of their excitement. Based on the biometrics obtained, the atmosphere of the venue and the behavior of the NPC audiences are changed, and interaction between the participants and the NPC audiences stimulates the three senses (visual, auditory, and haptic) to promote a sense of unity.

2.2 Moshing Function

The following is a description of the pseudo-moshing haptic feedback function that has been implemented. First of all, moshing is a violent clash of the participants that often occurs in live performances. To reproduce this unique behavior to live performances, participants wear a TactsuitX40 vest with 20 vibration devices attached to the front and 20 to the back, and Tactosy for Arms with 6 vibration devices attached to each arm. Past studies have shown that wearing the vibrating vest significantly increases embodiment and arousal in a VR environment [2]. These vibration devices can

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be used to generate a sustained vibration at the point of contact with the NPC audiences in the VR space, improving the NPC audiences' presence. Based on this function, surrounding NPC audiences will be brought closer to the participants at the most exciting points of the song, and by increasing the number of times they come into contact with each other, moshing will be recreated and a sense of unity will be promoted.

3 PRELIMINARY EXPERIMENT

In this section, we focus on our proposed moshing feature of Hype Live, and conduct a preliminary experimental investigation of its effectiveness. The timing of moshing was determined by obtaining the biometrics data of the two participants in advance.

3.1 Measurements

Our aim is to enhance the sense of unity and provide a better participation experience by interacting with NPC audience. Therefore, we conducted a preliminary experiment to test whether the proposed haptic feedback by vibration is effective or not. The sense of unity was measured by having the participants rate the four-item scale used by Yakura et al. [5] on a 7-point Likert scale (1: strongly disagree, 7: strongly agree). The questionnaires are as follows: 1) I feel like I was participating in the live concert with others. 2) I feel like many people were participating at the same time. 3) I feel like I was physically communicating with others. 4) I feel like I was rooting for the artist along with many others.

3.2 Conditions

This preliminary experiment was conducted in a within-participant design. Four participants joined the experiment: four men whose average ages were 23.25, and their S.D. was 0.957. The participants were divided into two groups. The participants in one group first watched a live performance using our system and vibration haptic feedback (Vibration condition), and then watched a similar live performance without vibration haptic feedback (Non-vibration condition). The participants in the other group watched the live performance in the non-vibration condition first, and then in the vibration condition.

3.3 Procedure

Before the experiment, the participants were given a brief explanation of its purpose and procedure. First, the participants were equipped with the necessary equipment and experienced a 30-second demonstration in each condition as a practice of watching a live performance. Next, the participants watched a full live performance of one song under the first condition and answered questionnaires to evaluate their impression of the live performance after watching it. Then, the participants watched the same live performance in a different condition and answered the same questionnaires. We used Unity-chan as the live performance artist. She is the 3D character model released by Unity Technologies Japan. We used her song "Unite in the Sky" and its corresponding dance choreography for this experiment. All participants watched the same pre-recorded performance of her in both conditions.

3.4 Results

Figure 3 shows the results of the questionnaires for each participant and averages. For all participants, we can see that the sense of unity is higher under vibration conditions than under non-vibration conditions.

4 DISCUSSION

4.1 Implications

After a preliminary experiment, one of the participants commented the vibration allowed him to feel the presence of people behind him without looking back. This result suggests that the proposed

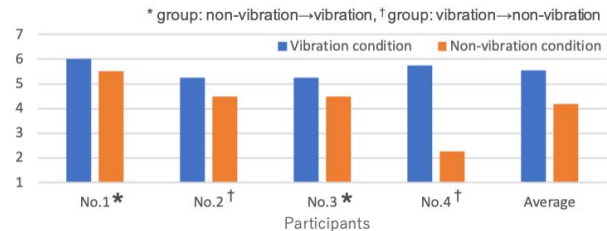


Figure 3: Degree of sense of unity

method has the potential to enhance the sense of unity. Therefore, this preliminary experiment shows that haptic feedback through vibration is an important interaction to create a sense of unity. As a future plan, we will improve the VR live venue and the behavior of NPC audiences, and conduct experiments with a large number of participants to prove that moshing with haptic feedback is one of the good methods to enhance the participants' sense of unity.

4.2 Limitations

As a limitation, we have reduced the number of NPC audiences to be placed in order to speed up the operation of the system. Also, this time, NPC audiences were moved towards the participants by moshing, but we are currently looking into the possibility of moving them in a disorganized manner. In the future, we will need to investigate the speed and timing of movements when moshing occurs and the number of NPC audiences.

5 CONCLUSION

In this study, we propose Hype Live, a system that improves the sense of unity by sharing the responses of the participants through visual, auditory, and haptic stimuli based on their biometrics. As a prototype of this system, we implemented a moshing behavior unique to live performances by providing participants with vibration feedback from surrounding NPC audiences using haptic devices. Preliminary experiments have shown that vibration feedback has the potential to improve the sense of unity. In the future, we plan to conduct experiments with a large number of participants to prove that simulated moshing can improve the sense of unity.

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